Chapter 10. Structure Learning: PC, GES and MMHC

Graphical models, Lecture 3 (constraint-based and score-based discovery of the Markov equivalence class)

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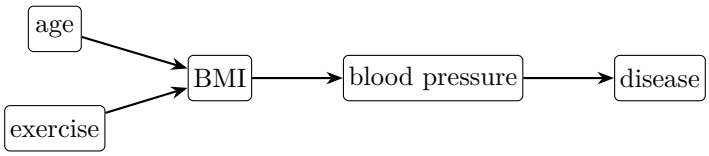
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1 What Observational Data Can Reveal

Questions of this lecture: which relations remain after accounting for the other variables? Which edges does conditional independence remove? How far can directions be fixed from observational data alone?

The toy example used throughout:



In a real analysis this graph is unknown and relations are searched for using observational data only. **The output of structure learning is the set of relations that the assumptions and the data allow; observational data do not fix every arrow.**

2 Assumptions Matter as Much as Data

Assumption	Practical meaning
acyclicity	relations at one time point are represented by a DAG without cycles
causal Markov condition	paths blocked in the graph appear as conditional independences in the data
faithfulness	no accidental cancellation of coefficients makes relations look independent
causal sufficiency	no unobserved common cause of two analyzed variables
same generating process	all samples come from the same system, without selection bias

PC and GES rely in particular on causal sufficiency. If hidden common causes are suspected, FCI is needed (Ch. 11).

3 The PC Algorithm

Motivation. Testing independence given every subset of variables is infeasible. The PC algorithm starts from the complete graph and removes edges using conditioning sets of increasing size taken only from current neighbours, then orients edges from the colliders it finds.

3.1 Skeleton phase

Algorithm: PC, skeleton

1. Connect all pairs of variables by undirected edges.
2. For $\ell = 0, 1, 2, \ldots$: for each adjacent pair (X_j, X_k) and each conditioning set $S \subseteq \text{adj}(X_j) \setminus \{X_k\}$ of size ℓ, test
$$H_0 : X_j \perp\!\!\!\perp X_k \mid X_S.$$
3. If H_0 is not rejected for some S, remove the edge $X_j - X_k$,
4. and store that S as the separating set $\text{Sep}(X_j, X_k)$.

Stop when no node has more than ℓ remaining neighbors.

The tests must suit the data: Gaussian continuous data use partial correlation with the Fisher-z test (Ch. 8); categorical data use the χ^2 or G^2 test; nonlinear relations need nonlinear CI tests (Ch. 11). The number of remaining edges depends on the significance level and sample size. The first output of PC is an undirected **skeleton**; an edge means a relation that remains after accounting for the other variables.

Example: skeleton of the toy example

Pair	Conditioning set	Decision	Action
age, exercise	$\varnothing$	independent	remove edge
age, blood pressure	BMI	conditionally independent	remove edge
exercise, blood pressure	BMI	conditionally independent	remove edge
BMI, disease	blood pressure	conditionally independent	remove edge

(Pairs such as age–disease and exercise–disease are removed similarly, e.g. given BMI.) The remaining skeleton is

$$\mathcal{E}_{\text{skel}} = \{\text{age–BMI},\ \text{exercise–BMI},\ \text{BMI–blood pressure},\ \text{blood pressure–disease}\}.$$

At this stage only the backbone of conditional relations is left; no direction has been decided yet.

3.2 Orientation phase

Colliders tell directions from observational data. The edge between age and exercise was removed with no conditioning set: $\text{Sep}(\text{age}, \text{exercise}) = \varnothing$. Both variables are adjacent to BMI, they are not adjacent to each other, and BMI is not in their separating set. So PC orients

$$\text{age} \to \text{BMI} \leftarrow \text{exercise}.$$

Before conditioning on BMI, age and exercise are independent; given BMI they can become dependent. If two non-adjacent variables point into the same middle variable, the middle variable is a collider.

Algorithm: PC, orientation

1. **v-structures**: for every unshielded triple $X - Z - Y$ (X, Y non-adjacent) with $Z \notin \text{Sep}(X, Y)$, orient $X \to Z \leftarrow Y$.
2. **Meek rules**, applied until nothing changes:

   R1 (no new collider) $X \to Z - Y$, X, Y non-adjacent $\Rightarrow Z \to Y$;

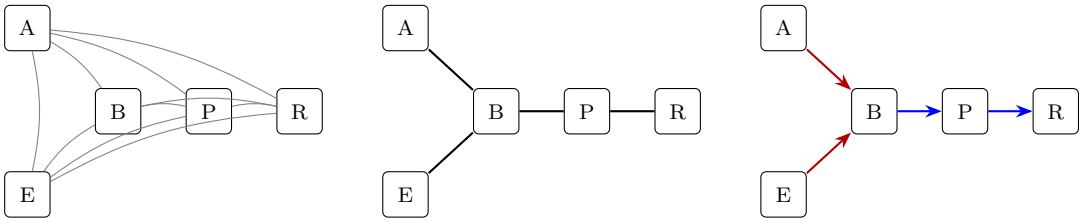
   R2 (no cycle) $X \to Z \to Y$ and $X - Y \Rightarrow X \to Y$;

   R3 $X - Z_1 \to Y$, $X - Z_2 \to Y$, $X - Y$, Z_1, Z_2 non-adjacent $\Rightarrow X \to Y$.

Preventing new colliders: if $A \to B - C$ and $A \not\sim C$, orienting $C \to B$ would create $A \to B \leftarrow C$, which the data did not support; hence $B \to C$. **Preventing cycles**: if $A \to B \to C$ and $A - C$, then $C \to A$ would create a directed cycle; hence $A \to C$. In the example the following directions are fixed in turn:

$$\text{age} \to \text{BMI} \leftarrow \text{exercise}, \qquad \text{BMI} \to \text{blood pressure} \to \text{disease}.$$

PC directions are interpreted only within the range forced by the conditional-independence pattern and the assumptions.



(1) complete graph (2) skeleton after CI tests (3) v-structure + Meek R1

Figure 1: PC on the toy example (A age, E exercise, B BMI, P blood pressure, R disease). Step 2 removes every edge whose endpoints become conditionally independent. Step 3 orients the v-structure $A \to B \leftarrow E$ (red, since $B \notin \mathrm{Sep}(A, E) = \varnothing$) and then propagates $B \to P \to R$ (blue) to avoid creating new colliders.

3.3 Why PC is correct

Theorem: correctness of PC with a CI oracle

Assume the Markov condition, faithfulness and causal sufficiency, and that every CI query is answered correctly. Then PC returns the CPDAG of the true DAG.

Proof [Supplement]. *Adjacencies.* An edge of G is never removed: by faithfulness, adjacent variables are dependent given every set. If X, Y are non-adjacent, one of them, say Y, is not a descendant of the other; by the local Markov property $X \perp\!\!\!\perp Y \mid \mathrm{pa}(X)$. Since true edges are never removed, $\mathrm{pa}(X) \subseteq \mathrm{adj}(X)$ at every stage, so this set (or another separating set) is found when $\ell = |\mathrm{pa}(X)|$. *v-structures.* Take an unshielded triple $X - Z - Y$. If Z is a non-collider on $X \cdot Z \cdot Y$, every set separating X, Y must contain Z (else that path is open); if Z is a collider, no separating set contains Z (conditioning on Z opens the path). So the single stored set $\mathrm{Sep}(X, Y)$ decides correctly. *Remaining edges.* The Meek rules are sound (each alternative orientation creates a new v-structure or a cycle) and complete for this situation (Meek, 1995), giving the CPDAG. ■

Remark: finite-sample issues

With real tests, an early type-II error removes an edge that later cannot be recovered and changes the separating sets used for orientation; errors propagate. The output also depends on the order in which pairs are tested; *PC-stable* removes this order dependence by updating adjacencies only after each level ℓ. Conflicting v-structures (both $X \to Z \leftarrow Y$ and $Z \to Y$ demanded) signal test errors or violated assumptions.

4 Score-Based Search

4.1 BIC for Gaussian linear SEMs

Score-based search compares candidate graphs by goodness of fit and complexity. For a candidate DAG G,

$$S_{\mathrm{BIC}}(G) = \ell(\hat{\theta}_G) - \frac{d_G}{2} \log n,$$

where $\ell(\hat{\theta}_G)$ measures how well the graph explains the data and d_G is the number of estimated parameters. Adding an edge can improve fit but increases the complexity penalty.

Theorem: BIC decomposes into local scores

For a Gaussian linear SEM,

$$S_{\mathrm{BIC}}(G) = \sum_{j=1}^{p} S_j\{X_j, \mathrm{pa}_G(j)\}, \qquad S_j = -\frac{n}{2}\log \hat{\sigma}^2_{j|\mathrm{pa}(j)} - \frac{|\mathrm{pa}(j)|+1}{2}\log n + c,$$

where $\hat{\sigma}^2_{j|\mathrm{pa}(j)}$ is the residual variance of the least-squares regression of X_j on its parents.

Proof [Supplement]. By the Markov factorization the log-likelihood is a sum of the log-likelihoods of the local regressions, whose parameters are separate. The maximized Gaussian log-likelihood of one regression is $-\frac{n}{2}\log(2\pi\hat{\sigma}^2) - \frac{n}{2}$, and that regression has $|\mathrm{pa}(j)|$ coefficients plus one variance (centered data). ■

Decomposability means that a local move (adding, deleting or reversing one edge) changes only one or two local scores, which makes search cheap. Markov-equivalent DAGs give the same maximum likelihood and the same number of parameters, so BIC is **score equivalent**: score-based methods also cannot distinguish directions within a Markov equivalence class from data alone.

4.2 GES and hill climbing

	GES	Hill climbing
search space	CPDAGs representing Markov equivalence classes	individual DAGs
moves	forward phase adds relations, backward phase removes them	add, delete or reverse an edge
typical output	a high-scoring CPDAG	one high-scoring DAG
caution	remaining undirected edges are undetermined	not every output direction is identified

Algorithm: Greedy Equivalence Search

Forward: start from the empty graph; repeatedly apply the single edge insertion (in equivalence-class space) that increases the score most, until no insertion helps. **Backward**: from there, repeatedly apply the best edge deletion until no deletion helps. Output the CPDAG.

Theorem: consistency of GES (Chickering, 2002)

If the distribution is faithful to a DAG and the score is consistent and decomposable (e.g. BIC), then GES returns the true CPDAG with probability tending to 1 as $n \to \infty$, despite being greedy.

Hill climbing over DAGs can get stuck in local optima; tabu lists and random restarts help. Computing a DAG as output and identifying arrows from data are two different things.

5 Computational Burden and Hybrid Methods

	Why the cost grows	Remaining problem
PC	repeats CI tests over many conditioning sets	tests multiply when the graph is dense; early test errors propagate to later edges and directions
GES	compares scores of many edge additions/deletions at each step	with many variables score computations grow; greedy results depend on sample and search path

For a maximal degree k, the number of PC tests is polynomial in p (roughly of order p^{k+2}), but it is exponential in k. Ways to reduce the burden:

- **MMHC**: reduce candidate edges by conditional-independence tests, then run a score search only within the restricted space;
- **FLOP**: search efficiently over variable orders and parent sets to cut the cost of score computations.

5.1 MMHC

$$\boxed{\text{CI tests}} \longrightarrow \boxed{\text{candidate skeleton}} \longrightarrow \boxed{\text{score search}} \longrightarrow \boxed{\text{DAG candidate}}$$

Stage 1 (candidate relations): for each variable, find the neighbors it may be directly connected to by CI tests (Max-Min Parents and Children); keep only those pairs as candidate edges. **Stage 2 (restricted score search)**: within the candidate edges, repeat additions, deletions and reversals, and select a DAG with high BIC or a similar score. An edge dropped in the first stage cannot be restored in the second, and not every direction of the output DAG is identified.

6 Comparison

	PC	GES	MMHC
approach	constraint (CI) based	score based	both, in sequence
search	removes edges when independent	changes the CPDAG to increase the score	restricts candidates by tests, then score search over DAGs
typical output	CPDAG	CPDAG	DAG candidate
key settings	independence test, significance level	score, complexity penalty	tests, significance level, score
sensitive to	test errors	score and search path	edges lost in stage 1

Summary

- Under acyclicity, Markov, faithfulness and causal sufficiency, observational data identify the Markov equivalence class.
- PC removes edges by CI tests, orients v-structures from the separating sets, then propagates orientations with the Meek rules; it is correct with a CI oracle.
- BIC decomposes into local regression scores and is score equivalent; GES searches equivalence classes and is consistent.
- MMHC combines both; its first-stage errors are irreversible.

7 Selected exercises

Motivation. The exercises check score equivalence and the consistency of BIC on a single edge, prove the soundness of the Meek rules used in the correctness proof of PC, and justify the polynomial bound on the number of tests.

Exercise 10.1 (BIC on one edge)

Two centered variables are observed n times, with sample variances s_X^2, s_Y^2 (divisor n) and sample correlation r, $|r| < 1$.
(a) Using the local scores of the BIC theorem, show that $S_{\mathrm{BIC}}(X \to Y) = S_{\mathrm{BIC}}(Y \to X)$ and that the edge beats the empty graph iff $-n \log(1 - r^2) > \log n$.
(b) For i.i.d. bivariate Gaussian data with correlation ρ, show that the probability that BIC chooses the correct adjacency tends to 1, both for $\rho = 0$ and for $\rho \neq 0$.

Solution [Supplement].

(a)

Step 1. The residual variance of Y on X is $s_Y^2 - s_{XY}^2/s_X^2 = s_Y^2(1 - r^2)$. Hence

$$S_{\mathrm{BIC}}(X \to Y) = -\frac{n}{2}\{\log s_X^2 + \log s_Y^2 + \log(1 - r^2)\} - \frac{3}{2}\log n + 2c,$$

which is symmetric in X and Y, so it equals $S_{\mathrm{BIC}}(Y \to X)$.

Step 2. The empty graph scores $-\frac{n}{2}(\log s_X^2 + \log s_Y^2) - \log n + 2c$. The difference is $-\frac{n}{2}\log(1 - r^2) - \frac{1}{2}\log n$, positive iff $-n \log(1 - r^2) > \log n$.

(b)

Step 3. ($\rho = 0$.) $\sqrt{n}\, r \xrightarrow{d} N(0, 1)$ (the t or Fisher-z results of Ch. 8 with $q = 0$), so nr^2 is bounded in probability and $r^2 \xrightarrow{p} 0$. For $0 \le u \le \frac{1}{2}$, $-\log(1 - u) = \sum_{k \ge 1} u^k/k \le u/(1 - u) \le 2u$. On the event $\{r^2 \le \frac{1}{2}\}$, whose probability tends to 1,

$$P\{-n \log(1 - r^2) > \log n\} \le P\{2nr^2 > \log n\} + P\{r^2 > \tfrac{1}{2}\} \to 0,$$

because $\log n \to \infty$ while nr^2 stays bounded in probability.

Step 4. ($\rho \neq 0$.) $r \xrightarrow{p} \rho$, so $-\log(1 - r^2) \xrightarrow{p} -\log(1 - \rho^2) > 0$, and $-n \log(1 - r^2)/\log n \xrightarrow{p} \infty$ because $n/\log n \to \infty$. Hence the edge is chosen with probability tending to 1.

The penalty $\frac{1}{2}\log n$ grows, so null partial correlations of size $O_p(n^{-1/2})$ are eventually rejected, yet slowly enough that fixed nonzero correlations are always kept. ■

Exercise 10.2 (Soundness of the Meek rules)

Let $\mathcal{C}$ be a Markov equivalence class and P a partially directed graph with the common skeleton such that every directed edge of P has that orientation in every DAG of $\mathcal{C}$, and every v-structure of $\mathcal{C}$ appears in P with both edges directed. Show that if the premise of R1, R2 or R3 holds in P, then every DAG of $\mathcal{C}$ contains the edge that the rule orients.

Solution [Supplement]. Fix $G' \in \mathcal{C}$. The two variables in each conclusion are adjacent, so G' orients their edge one way or the other; we exclude the wrong way.

Step 1. (R1: $X \to Z - Y$, X, Y non-adjacent.) G' contains $X \to Z$. If it contained $Y \to Z$, then $X \to Z \leftarrow Y$ would be a v-structure of G', hence of $\mathcal{C}$, and would appear in P with $Y \to Z$ directed; but $Z - Y$ is undirected in P. So G' contains $Z \to Y$.

Step 2. (R2: $X \to Z \to Y$, $X - Y$.) If G' contained $Y \to X$, then $X \to Z \to Y \to X$ would be a directed cycle. So G' contains $X \to Y$.

Step 3. (R3: $X - Z_1 \to Y$, $X - Z_2 \to Y$, $X - Y$, Z_1, Z_2 non-adjacent.) Suppose G' contains $Y \to X$. For $i = 1, 2$, $X \to Z_i$ would create the cycle $X \to Z_i \to Y \to X$, so G' contains $Z_i \to X$. Then $Z_1 \to X \leftarrow Z_2$ with Z_1, Z_2 non-adjacent is a v-structure of $\mathcal{C}$ and must appear directed in P; but $X - Z_1$ is undirected in P. So G' contains $X \to Y$.

After the v-structure step of PC (with a CI oracle) the graph satisfies the assumptions on P, and each orientation keeps them, so all directions added by the Meek rules are correct. ■

Exercise 10.3 (How many tests PC needs)

Assume the conditions of the PC correctness theorem and let every node of the true DAG have at most k neighbours, $p \geq 3$. Show that the skeleton phase runs only the levels $\ell = 0, \ldots, k$ and performs at most $(k+1)p^{k+2}$ CI tests.

Solution [Supplement].

Step 1. By the proof of the correctness theorem, a non-adjacent pair X, Y (with Y not a descendant of X) is separated by $\mathrm{pa}(X)$, so its edge is removed at the latest at level $\ell = |\mathrm{pa}(X)| \leq k$; true edges are never removed. So after level k the current graph is the true skeleton.

Step 2. At that point no node has more than k neighbours, so the stopping rule ends the skeleton phase; levels $\ell \geq k+1$ are never run.

Step 3. At level ℓ, an ordered adjacent pair (X_a, X_b) is tested with at most $\binom{|\mathrm{adj}(X_a)|-1}{\ell} \leq \binom{p-2}{\ell} \leq (p-2)^\ell$ sets, and there are at most $p(p-1)$ ordered pairs. Hence the total is at most

$$p(p-1)\sum_{\ell=0}^{k}(p-2)^\ell \leq p^2(k+1)(p-2)^k \leq (k+1)p^{k+2}.$$

The bound is polynomial in p for fixed sparsity, but the exponent grows with the maximal degree: dense graphs make PC expensive, and they also require large conditioning sets, for which the tests lose power. ■